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comparative evaluation, exceptional performances, attribution dilution, finite attention, substitutability, organizational risk, opportunity suppression, selection scarcity, capability, sustainable substitutability, congestion crowding, elite environments, identifiability parameters, economic variables, trade-offs

SPEAKERS

Alex Gates, Charles Levine

Charles Levine 00:00

Okay, so first you talked about the different, the different classes, right? So comparative evaluation, the first thing, right? In other words, these are mechanisms. I couldn't help but notice your second bullet said stand-out performances appear less exceptional.

Alex Gates 00:18

Yes,

Charles Levine 00:18

ironically, what we're really saying is exceptional performances appear less stand out, isn't that wild? That's right.

Alex Gates 00:27

Yes, I like that.

Charles Levine 00:29

Then you talked about attribution, dilution, love it, finite attention, and by the way, these are things that my earlier model that I started down the road with had variables, some of them in there, right, which is good, because we're thinking the same thing. Borderline elite candidates are substitutable. I love this concept of substitutability.

Alex Gates 00:53

Yes,

Charles Levine 00:53

right. And then the first thing that I thought was the substitutable borderline candidates, those are the people, just the cut line of the NIH in Wang's paper,

Alex Gates 01:05

they are

Charles Levine 01:06

right, boom, boom, boom, and then, but I love that after our discussion, you brought in organizational risk minimization,

Alex Gates 01:15

yes,

Charles Levine 01:15

right, that's money ball, yes, I couldn't help but write this in NBA. What you're basically saying is then you might fall back on heuristics, which is like voodoo. Is that another way of saying like voodoo selection? Or if several team minutes look promising, Scott's may prefer the more physically prototypical younger already famous player, and I added or white, well, but that's just me now. Opportunity suppression, great. I love the mechanism becomes recursive, brilliant. I would not have thought of that word, it's perfect now. Qing and timing, is this just another way of saying selection scarcity?

Alex Gates 02:02

Yes. Okay,

Charles Levine 02:03

but breaking it down into both the temporal aspect,

Alex Gates 02:08

yes,

Charles Levine 02:08

or a temporal aspect.

Alex Gates 02:10

This is there are many different ways that scarcity can be created,

Charles Levine 02:14

right?

Alex Gates 02:15

Right, and some of those are about time, some of those are about comparison, some of those are about just finite.

Charles Levine 02:25

Yeah, I think it's a couple of these things that seemed a little

Alex Gates 02:29

okay. Don't get too deep into those weeks, okay? They're more like these are different facets and reasons to consider why this is important to include,

Charles Levine 02:41

yeah,

Alex Gates 02:41

and we should address some of it, yeah, not all of it, and get your head going.

Charles Levine 02:46

Great, I got it. Capability and sustainable substitutability. I could not, for the life of me, figure out what you meant by capability.

Alex Gates 03:00

So that's here after where

Charles Levine 03:06

we say after it's in sixth is queuing and timing,

Alex Gates 03:09

yeah, sixth is queuing and timing, right. So then

Charles Levine 03:15

the strongest theoretical framework is probably elite environments increase both capability and sustainability. What do we mean by capability?

Alex Gates 03:28

It's probably.. it's a good question.

Charles Levine 03:34

You can say I'm not sure, because it has to be constantly. At first, I thought, does that mean developmental?

Alex Gates 03:39

Yeah,

Charles Levine 03:40

is it development, or is it, or is it the organizational capability?

Alex Gates 03:46

I think it should be organizational capability,

Charles Levine 03:49

because it said weak environments have low capability, low suitability, moderately strong environments have high capability.

Alex Gates 03:56

Yeah,

Charles Levine 03:57

but manageable, and then elite environments, so all I could think was the capability is referring to organizational capability, tried plugging in developmental, and that doesn't work either,

Alex Gates 04:17

I think. No, it should be something about own performance and how differentiated an individual is from their environment. I think that's capability should be like own performance of the top people, right? If you think about the difference here, is it's not just about everybody, it's like who are the best people, right? So a moderate, strong environment should have still have very good people in it, but they stand out. There's there's much less substitutability. They stand out. It's easy to identify who they are, but in elite environments, you know, the top people are extremely high, but they're not. There's a lot of substitutability, everybody in them is good.

Charles Levine 05:04

Okay, so we think that's really just that's a proxy for latent talent.

Alex Gates 05:09

Yeah, I would, I would put that there. Okay,

Charles Levine 05:13

and I discovered that what I was telling you about, summing up the was is mathematically the same as the mean, because the teams are the same size, and so, yeah, it was those ones who was anything else. Oh, the theory predicts elite superstars still get drafted, that's that's the bump, right? That little bump we saw at the very end.

Alex Gates 05:46

Oh, really?

Charles Levine 05:47

Yeah, where you said,

Alex Gates 05:48

okay,

Charles Levine 05:48

even stronger. Show that downturn exists primarily. The first thing it said was elite superstars will still get drafted.

Alex Gates 05:55

Yeah,

Charles Levine 05:56

there's the proof.

Alex Gates 05:57

They still get up there in the very best teams,

Charles Levine 06:01

okay? Oh. viable pier density mediates the lean. I like this congestion, should delay success, not only reduce it, which is in the Wang model, is borne out, but the existing members, this got like, I'm not going to get too much attention to it, but that's almost like, hey, if you're a new subscriber, we'll give you the deal, but if you're existing subscriber, we won't. Okay,

Alex Gates 06:44

so what I think are important features to take away is that second set of things, it's suggesting new features to be calculating out of the data,

Charles Levine 06:58

right?

Alex Gates 06:58

Right. So, for example, coming up with measures of congestion, like, and explicitly quantify what those are,

Charles Levine 07:07

right,

Alex Gates 07:08

in the MBA, for example, because you have the raw data available the

Charles Levine 07:12

way that I had. Yeah, so I have, in the beginning, I had congestion quality, then I also had congestion crowding,

Alex Gates 07:19

right,

Charles Levine 07:19

and I think the crowding is what we really need to tease,

Alex Gates 07:21

that's what it is here. Yep, so there are some ways in calculating that, right? They're actually pretty easy. It's just like how many people are above the threshold, and that's an easy, easy kind of calculation.

Charles Levine 07:37

And then take a small fraction of that and subtract it from the the innate signal, the a sub I,

Alex Gates 07:42

yeah. And then I like some of their natural experiments that they were trying to get at, right? Kentucky Duke mega recruiting years, yeah. Like, what happens there?

Charles Levine 08:00

Yeah. it isn't until the beginning of our conversation that I realized that you didn't write this entire thing yourself. Okay, good. I feel much better.

Alex Gates 08:13

Yeah, no, no, this was done in 45 minutes, but

Charles Levine 08:17

great, which is as it should be, right? Yeah, we have the tools, let's use them,

Alex Gates 08:20

right?

Charles Levine 08:22

Yeah, okay. So that's really what I'm going to do now, is just focus on that, adding that congestion crowding term.

Alex Gates 08:28

Yeah, so you see the model that I gave you the code for,

Charles Levine 08:31

I've already got it put into Jupiter

Alex Gates 08:33

Notebook. Great, so the big questions are, How identifiable are some of those parameters from data you have real world data, right? You even fit this model to to the three worlds that we have the data for at the moment, or are there identifiability issues? Right, so none of that's worked out very well here. I didn't do any of the explorations of the extreme events, so you should do some, you know, I found just a nice parameter combination that kind of produces the effect that we want, but

Charles Levine 09:12

what do you mean by extreme events?

Alex Gates 09:13

Turn stuff on and off, oh, you know, make some of the parameters go to zero or run, or whatever it is, right? So validate that a lot of that still exists in this model, so identifiability parameters, then the some of the track down some of the things it suggests to look at, especially the congestion stuff, and see how hard that is to calculate a version of that in the MBA data, and what else can you get along those lines. I just did one or two quick prompts about that, but there probably is a lot. More, we could.

Charles Levine 10:01

I'm not sure I understand the third thing. How is that different from how identifiable?

Alex Gates 10:06

So, identifiable is let's fit this model, which has what, five or six parameters, to the data, and is it possible to come up with a unique set of parameters that we produces

Charles Levine 10:18

that we can map to

Alex Gates 10:19

the data to the

Charles Levine 10:20

tables,

Alex Gates 10:20

or no, that are that are recovered as uniquely defining what's in the data, right? That's what it means to be identifiable in this sense. Okay, we fit the model to our data.

Charles Levine 10:32

Yeah,

Alex Gates 10:32

right. I think it's over parameterized. I think there's probably too many different things we could be moving, and it's not going to recover exactly, but I don't know, so then if it doesn't recover exactly, or if it doesn't recover, and it's there's like lots of parameter combinations that could work, make some reasonable fixes, and try to lower the dimensionality down,

Charles Levine 10:55

okay?

Alex Gates 10:56

So that you can still fit the three different data sets that we have,

Charles Levine 10:59

okay,

Alex Gates 11:00

in a meaningful way,

Charles Levine 11:00

notice you're saying create variable y , which in this data set we roll a , b , and c into. In this data set we roll a , b , and d into.

Alex Gates 11:09

Yeah, right. Or fix for everyone, where you drawing from the same district beta distribution with this parameter here. And once we fix that,

Charles Levine 11:19

okay,

Alex Gates 11:19

we're not fitting that from data, yeah, you know, it's like something

Charles Levine 11:23

like something, it's same across the board,

Alex Gates 11:25

yeah.

Charles Levine 11:26

Okay,

Alex Gates 11:26

but the third thing I was talking about was this idea that the model should suggest new things to look at or focus on in our data sets, right?

Charles Levine 11:41

Right,

Alex Gates 11:41

congestion. as an explicit object is one of them. Now we're going to be exploring, can you quantify the extent of congestion, and are there certain properties that that obeys, which were not apparent to us had we not decided that that was an important component of our model? Right? Just proposed,

Charles Levine 12:01

yeah,

Alex Gates 12:02

right. What else? What else does the model suggest we should be looking at as relevant features of the system that we could then go and explore empirically in the data sets there? I

Charles Levine 12:15

feel like ideally we should be able to find some variable like congestion that indirectly contains individual talent, but does not directly contain it. So, we could look at a system and just measure whatever that congestion quantity

Alex Gates 12:37

is,

Charles Levine 12:38

and predict how it's going to affect the people in the organization?

Alex Gates 12:42

Yes, right, yep. So those are the big priorities. A fourth priority to let seep over the next few weeks. All three of the systems we've identified, we carefully picked because we thought that they met all the conditions that now give us the effect that we want.

Charles Levine 13:02

Right?

Alex Gates 13:03

Can you identify a fourth or fifth system based on what you now know, and maybe how the model posits the system should be that break it, so the effect is not there in the system that you're identifying. Hm. Do you understand what I'm, yeah,

Charles Levine 13:24

you're saying. Well, let's take.. if we take a look at something else, it turns out that

Alex Gates 13:33

NHL draft..

Charles Levine 13:35

it turns out that this doesn't take place at all, right? Right, that there's no downturn,

Alex Gates 13:41

that there's no

Charles Levine 13:42

doubt, yep,

Alex Gates 13:43

yeah, and it's for an obvious reason that's breaking something key to our model type of thing,

Charles Levine 13:52

and we're doing that because we want to better understand our model, but also because a good paper also says these are the downsides, these are the problems with my,

Alex Gates 14:01

yeah, if your model fit everything, it would be much more circumspect than if there's clear limitations on what it clear cannot explain. Okay, so think about that, and likewise, if there are other systems that you can now think of that we could get data for. I

Charles Levine 14:24

sort of thought you were going, but

Alex Gates 14:25

yeah, so that's another direction. A clear question here is, there's trade-offs in the importance of some of these terms, and so do we see in these systems if we measure the term, we could expect this steep of a decline, this positioning of the, you know, where and how some of the features materialize,

Charles Levine 14:49

right?

Alex Gates 14:49

We could even predict that based on some simple calculation of the data, rather than going through the whole..

Charles Levine 14:56

my gut tells me what we're looking for is. Something to do with economics,

Alex Gates 15:02

something,

Charles Levine 15:03

something like GDP of a nation versus or patent applications, or,

Alex Gates 15:13

oh, interesting, yeah, you know what I'm saying, yep,

Charles Levine 15:16

something, where, or we could say that, you know, how many million barrels of oil did this country in the Middle East produce versus its gross domestication? We might find something like that.

Alex Gates 15:30

Okay,

Charles Levine 15:32

but something economic, I think it'll be easy to get in terms of nations

Alex Gates 15:36

competing,

Charles Levine 15:37

you know, in terms of nations competing for

Alex Gates 15:39

companies,

Charles Levine 15:40

for companies, whatever success

Alex Gates 15:42

is really nice level of estimation. Yeah, great, you have the right idea.

Charles Levine 15:48

Thanks.